

Robert Williams

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EDUCATION

New Jersey Institute of Technology (NJIT) | Newark, NJ

Expected May 2027

Bachelor of Science in Computer Science | Minor in Applied Mathematics | GPA: 3.4/4.0

Relevant Coursework: Data Structures & Algorithms, Intro to Artificial Intelligence, Linear Algebra, Database Systems, Computer Networks, Stochastic Processes, Intensive Programming in Linux

PROJECTS

ML Model Registry & Deployment Platform [GitHub]

Python, FastAPI, PostgreSQL, SQLAlchemy, AWS S3, Docker

- Implemented model registry service with FastAPI and SQLAlchemy supporting versioned model entities, metadata schemas, and state transitions (staging → production) backed by PostgreSQL
- Integrated S3 artifact storage with pre-signed URLs and SSE-KMS encryption; containerized the full application with Docker for reproducible local development and deployment
- Built promotion and rollback endpoints with Pydantic request/response validation and CRUD operations; verified API behavior through systematic endpoint testing with Postman and FastAPI's interactive docs

Time-Series Air Quality Forecasting System [GitHub]

Python, TensorFlow/Keras, Pandas, scikit-learn, Matplotlib

- Developed multi-output LSTM neural network to forecast 6 air pollutants across Seoul monitoring stations, achieving R² scores of 0.93 (SO₂), 0.90 (NO₂), and 0.86 (O₃) on 647K+ temporal sequences
- Engineered data preprocessing pipeline with temporal train/test splitting, outlier detection (Z-score, IQR), and MinMaxScaler fitted exclusively on training data, eliminating data leakage

Plant Disease Classification System [GitHub]

Python, PyTorch, torchvision, ResNet-18

- Fine-tuned ResNet-18 via transfer learning on 108K PlantVillage images, replacing the final FC layer (512→38 classes) to classify diseases across 14 crop species with 98.64% test accuracy
- Implemented data augmentation pipeline with random cropping and ImageNet normalization, training for 10 epochs to achieve robust generalization across 38 disease categories

EXPERIENCE

Rutgers University – Newark | Newark, NJ

June 2025 – September 2025

Business Clerical Assistant, Student Affairs Business Office

- Processed financial transactions and business expenses for 15+ Student Affairs departments using university financial management systems, ensuring compliance with institutional accounting policies
- Managed vendor invitations, payment processing, and budget reconciliation activities, maintaining 99%+ accuracy in financial documentation and transaction records
- Served as front-line support for departmental inquiries regarding financial policies, troubleshooting accounting system issues and providing procedural guidance to staff and stakeholders

The Greenwich Group | New York, NY

June 2023 – September 2023

Strategic Business Development and Innovation Intern

- Analyzed partnership opportunities across biotech, lending, and real estate sectors using Python (Pandas, NumPy) and Excel, delivering data-driven insights to executive leadership
- Researched IoT adoption trends and productivity impact across 50+ enterprise case studies, synthesizing findings into actionable reports for senior management

LEADERSHIP & ACTIVITIES

National Society of Black Engineers (NSBE) | NJIT Chapter

September 2023 – Present

Academic Excellence Committee Member

- Led community outreach initiatives leveraging social media campaigns, driving 27% increase in chapter engagement and expanding corporate partnerships with Amazon, Kearfott, and Emma Bowen Foundation
- Coordinate technical workshop series connecting 100+ members with industry professionals for career development and networking

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript, SQL

Frameworks & Libraries: FastAPI, Flask, TensorFlow, PyTorch, scikit-learn, SQLAlchemy, Pydantic, Pandas, NumPy, Node.js, Express.js, React.js

DevOps & Cloud: Docker, AWS (S3, EC2, KMS, IAM), Git, GitHub, Linux/Unix, PostgreSQL, MongoDB

Tools: Postman, Matplotlib, Seaborn, Jupyter Notebook